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FOLDING BASKET

Abstract

A folding basket includes a bottom frame panel having a first bottom frame and a second bottom frame on each long and short sides. A first side frame panel is pivoted to the first bottom frame. A second side frame panel is pivoted to the second bottom frame. Two accommodating portions are horizontally recessed in an outer side of the first side frame panel. A locking member is horizontally and movably installed in each accommodating portion. Two ends of the locking members are respectively engaged with side edges of the first and second side frame panels. When the locking members are pulled, the locking relationship between the two ends of the locking members and the side edges of the first sand second ide frame panels is released, so that the first side frame panel and the second side frame panel are foldable inward sequentially.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to a basket, and more particularly, to a folding basket capable of installed to a bicycle rear rack.

BACKGROUND OF THE INVENTION

[0002] The conventional storage baskets are mostly one-piece molded and fixed in shape. When not containing items, this type of storage basket occupies a large volume due to its inability to fold, and may cause storage problems. In order to improve the above shortcomings, foldable storage baskets are also available on the market to meet different needs.

[0003] The present invention intends to provide a foldable basket to eliminate the shortcomings mentioned above.

SUMMARY OF THE INVENTION

[0004] The present invention relates to a folding basket and comprises a bottom frame panel which includes a first bottom frame on each long sides, and a second bottom frame on each short sides. Each first bottom frame has multiple first pivot connectors, and each second bottom frame has multiple second pivot connectors. A first side frame panel is pivoted to the first bottom frame. The first side frame panel has multiple first pivot parts located at a bottom thereof and corresponding to the multiple first pivot connectors. A second side frame panel is pivoted to the second bottom frame. The second side frame panel has multiple second pivot parts located at a bottom thereof and corresponding to the multiple second pivot connectors. Two accommodating portions are horizontally recessed in an outer side of the first side frame panel. A locking member is horizontally and movably installed in each accommodating portion. Two ends of the locking members are respectively engaged with side edges of the first side frame panel and the second side frame panel. When the locking members are pulled, the locking relationship between the two ends of the locking members and the side edges of the first side frame panel and the second side frame panel is released so that the first side frame panel and the second side frame panel are foldable inward sequentially.

[0005] The present invention will become more obvious from the following description when taken in connection with the accompanying drawings which show, for purposes of illustration only, a preferred embodiment in accordance with the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a perspective view of the folding basket of the present invention;

[0007] FIG. **2** is another perspective view of the folding basket of the present invention;

[0008] FIG. **3** is an enlarged view of the circled “A” of FIG. **1**;

[0009] FIG. **4** is an enlarged view of the circled “B” of FIG. **1**;

[0010] FIG. **5** is a perspective view of the locking member of the present invention;

[0011] FIG. **6** is another perspective view of the locking member of the present invention;

[0012] FIG. **7** is a plan view of the locking member of the present invention;

[0013] FIG. **8** is a schematic diagram to show that the folding basket is installed on a bicycle rear rack;

[0014] FIG. **9** is a side view plan of the action of the locking member of the present invention;

[0015] FIG. **10** is another side view plan of the action of the locking member of the present invention;

[0016] FIG. **11** is a sectional view, taken along line VIII-VIII of FIG. **1**;

[0017] FIG. **12** is an enlarged view of the circled “C” of FIG. **11** of the present invention;

[0018] FIG. **13** is a schematic diagram to show the folding of the first side frame panel of the present invention, and

[0019] FIG. **14** is a schematic diagram to show the folding of the second side frame panel of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0020] Referring to FIGS. **1** to **14**, the folding basket **5** of the present invention comprises a bottom frame panel **1** having a first bottom frame **10** on each long sides, and a second bottom frame **11** on each short sides. Each first bottom frame **10** has multiple first pivot connectors **12**, and each second bottom frame **11** has multiple second pivot connectors **13**.

[0021] A first side frame panel **2** is pivoted to the first bottom frame **10**. The first side frame panel **2** has multiple first pivot parts **21** located at a bottom thereof and corresponding to the multiple first pivot connectors **12**. The first side frame panel **2** includes a gripping recess **20** for convenience of gripping the folding basket **5**. The first side frame panel **2** has first engaging parts **22** and first receiving surfaces **23** at side edges thereof. Two accommodating portions **24** are horizontally recessed in an outer side of the first side frame panel **2**. Each of the two accommodating portions **24** on the outer side of the first side frame panel **2** has a limiting plate **240**, a first blocking plate **241** and a second blocking plate **242**. A first storage space **243** is formed between the limiting plate **240** and the first blocking plate **241**, and a second storage space **244** is formed between the limiting plate **240** and the second blocking plate **242**. A positioning hole **2430** is defined in an inside wall of the first storage space **243**. The limiting plate **240** has a first perforation **2400**, and the second blocking plate **242** has a second perforation **2420**. In this embodiment, one of two side end faces of the first side frame panel **2** is a first planar segment **25**, and another one of the two side end faces slopes downward from the top to form an inclined surface segment **26**, then extends downward to form a second planar segment **27**.

[0022] A second side frame panel **3** is pivoted to the second bottom frame **11**. The second side frame panel **3** has second engaging parts **31** which are located corresponding to the first engaging parts **22**. The second side frame panel **3** also has a second receiving surfaces **32** which is adjacent to the first receiving surfaces **23**. In this embodiment, both side end faces of the second side frame panel **3** have first bending segments **33** and second bending segments **34** corresponding to the first planar segment **25** and the second planar segment **27** of the first side frame panel **2** respectively.

[0023] A locking member **4** is horizontally and movably installed in each accommodating portion **24**. Each of the locking members **4** has a rod body **40**, a handle portion **41** and a locking portion **42** are respectively formed at both ends of the rod body **40**. The handle portion **41** has a positioning part **410** facing the accommodating portion **24**, and the locking portion **42** includes a hollow groove **420** facing the accommodating portion **24**. An elastic limiting portion **43** is located between the handle portion **41** and the locking portion **42**. The elastic limiting portion **43** of the locking member **4** has two elastic arc pieces **430** respectively set on upper and lower sides of the rod body **40**.

[0024] When assembling, the multiple first pivot parts **21** of the first side frame panel **2** and the multiple second pivot parts **30** of the second side frame panel **3** are respectively pivoted to the multiple first pivot connectors **12** of the first bottom frame **10** are respectively pivotably connected to the first pivot connectors **12** of the first bottom frame **10** and the second pivot connectors **13** of the second bottom frame **11**. Accordingly, the first side frame panel **2** and the second side frame panel **3** can be fold inward toward the central position of the bottom frame panel **1**. Since the first engaging part **22** of the first side frame panel **2** is connected to the second engaging part **31** of the second side frame panel **3** from the inner side of the second side frame panel **3** to the outer side of the second side frame panel **3** (as shown in FIGS. **3** and **4**), so that the second side frame panel **3** must be folded after the first side frame panel **2** is folded inward.

[0025] Subsequently, the rod body **40** extends through the first perforation **2400** of the limiting plate **240**, causing the elastic limiting portion **43** to be constrained by the limiting plate **240** in the second storage space **244**. Meanwhile, the handle portion **41** is positioned in the first storage space **243**, which has positioning holes **2430** corresponding to the positioning parts **410** of the handle portion **41**. The positioning relationship between the positioning parts **410** and the positioning holes **2430** enhances the stability of the handle portion **41** in the first storage space **243**. The locking portion **42** passes through the second perforation **2420** of the second blocking plate **242** and forms a locking relationship with the first receiving surface **23** of the first side frame panel **2** and the second receiving surface **32** of the second side frame panel **3**.

[0026] When in use, as shown in FIGS. **8** and **14**, the folding basket **5** of the present invention is mounted on a bicycle rear rack **6** by using assembly components **50**, and items are placed inside the folding basket **5**. The gripping recesses **20** of the first side frame panel **2** facilitate users in moving the folding basket **5**. When the folding basket **5** does not need to hold items, users can laterally pull the locking member **4** (as shown in FIGS. **9** and **10**). During the process of the rod body **40** passing through the first perforation **2400** of the limiting plate **240**, the two elastic arc pieces **430** on the upper and lower sides of the rod body **40** are compressed elastically. After passing through the first perforation **2400**, the two elastic arc pieces **430** are limited in the first storage space **243** due to elastic recovery. At the same time, after the locking portion **42** of the rod body **40** passes through the second perforation **2420** of the second blocking plate **242**, the locking relationship between the locking portion **42** and the first receiving surface **23** of the first side frame panel **2** and the second receiving surface **32** of the second side frame panel **3** can be released. Subsequently, the first side frame panel **2** is folded inward (as shown in FIG. **13**), and after the first side frame panel **2** is folded inward, the engagement relationship between the first engaging part **22** of the first side frame panel **2** and the second engaging part **31** of the second side frame panel **3** is released, allowing the second side frame panel **3** to be folded inward (as shown in FIG. **14**). In other words, by controlling whether the locking member **4** is pulled, the locking relationship between the locking member **4** and the first side frame panel **2** and the second side frame panel **3** is established or released, achieving the sequential inward folding of the first side frame panel **2** and the second side frame panel **3**, completing the storage action.

[0027] It is noted that the locking member **4** has a hollow groove **420** facing the accommodating portion **24**, which allows for a gap between the locking member **4** and the accommodating portion **24**, avoiding excessive tightness and friction in the connection between them and improving the smoothness of pulling the locking member **4**.

[0028] While we have shown and described the embodiment in accordance with the present invention, it should be clear to those skilled in the art that further embodiments may be made without departing from the scope of the present invention.

Claims

1. A folding basket (**5**) comprises: a bottom frame panel (**1**) having a first bottom frame (**10**) on each long sides, and a second bottom frame (**11**) on each short sides, each first bottom frame (**10**) having multiple first pivot connectors (**12**), each second bottom frame (**11**) having multiple second pivot connectors (**13**); a first side frame panel (**2**) pivoted to the first bottom frame (**10**), the first side frame panel (**2**) having multiple first pivot parts (**21**) located at a bottom thereof and corresponding to the multiple first pivot connectors (**12**); a second side frame panel (**3**) pivoted to the second bottom frame (**11**), the second side frame panel (**3**) having multiple second pivot parts (**30**) located at a bottom thereof and corresponding to the multiple second pivot connectors (**13**), wherein two accommodating portions (**24**) are horizontally recessed in an outer side of the first side frame panel (**2**), a locking member (**4**) is horizontally and movably installed in each accommodating portion (**24**), two ends of the locking members (**4**) are respectively engaged with side edges of the first side frame panel (**2**) and the second side frame panel (**3**), when the locking members (**4**) are pulled, the locking relationship between the two ends of the locking members (**4**) and the side edges of the first side frame panel (**2**) and the second side frame panel (**3**) is released so that the first side frame panel (**2**) and the second side frame panel (**3**) are foldable inward sequentially.

2. The folding basket as claimed in claim 1, wherein the side edges of the first side frame panel (**2**) have first engaging parts (**22**), the second side frame panel (**3**) has second engaging parts (**31**) which are located corresponding to the first engaging parts (**22**).

3. The folding basket as claimed in claim 1, wherein each of the two accommodating portions (**24**) on the outer side of the first side frame panel (**2**) has a limiting plate (**240**), a first blocking plate (**241**) and a second blocking plate (**242**), a first storage space (**243**) is formed between the limiting plate (**240**) and the first blocking plate (**241**), a second storage space (**244**) is formed between the limiting plate (**240**) and the second blocking plate (**242**), the limiting plate (**240**) has a first perforation (**2400**), the second blocking plate (**242**) has a second perforation (**2420**).

4. The folding basket as claimed in claim 3, wherein the side edges of the first side frame panel (**2**) has first receiving surfaces (**23**), the second side frame panel (**3**) has a second receiving surfaces (**32**) which is adjacent to the first receiving surfaces (**23**), each of the locking members (**4**) has a rod body (**40**), a handle portion (**41**) and a locking portion (**42**) are formed at both ends of the rod body (**40**), an elastic limiting portion (**43**) is located between the handle portion (**41**) and the locking portion (**42**), the handle portion (**41**) is located in the first storage space (**243**) corresponding thereto, the elastic limiting portion (**43**) is located in the second storage space (**244**) corresponding thereto, the locking portions (**42**) are engaged with the first receiving surfaces (**23**) and the second receiving surfaces (**32**) respectively to form a locking relationship.

5. The folding basket as claimed in claim 4, wherein the elastic limiting portion (**43**) of the locking member (**4**) has two elastic arc pieces (**430**) respectively set on upper and lower sides of the rod body (**40**).

6. The folding basket as claimed in claim 4, wherein the handle portion (**41**) has a positioning part (**410**) which is located corresponding to a positioning hole (**2430**) defined in an inside wall of the first storage space (**243**).

7. The folding basket as claimed in claim 4, wherein the locking portion (**42**) includes a hollow groove (**420**) facing the accommodating portion (**24**), the hollow groove (**420**) avoids friction between the locking member (**4**) and the accommodating portion (**24**) to improve smoothness of pulling the locking member (**4**).

8. The folding basket as claimed in claim 1, wherein one of two side end faces of the first side frame panel (**2**) is a first planar segment (**25**), another one of the two side end faces slopes downward from the top to form an inclined surface segment (**26**), then extends downward to form a second planar segment (**27**).

9. The folding basket as claimed in claim 7, wherein both side end faces of the second side frame panel (**3**) have first bending segments (**33**) and second bending segments (**34**) corresponding to the first planar segment (**25**) and the second planar segment (**27**) of the first side frame panel (**2**) respectively.

10. The folding basket as claimed in claim 1, wherein the first side frame panel (**2**) has a gripping recess (**20**).

11. The folding basket as claimed in claim 1, wherein the folding basket (**5**) is mounted on a bicycle rear rack (**6**) using assembly components (**50**).